

CHAPTER 13 – Automated Assessment of Learner-Generated Natural Language Responses

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Introduction

Eliciting open-ended learner responses gives learners the opportunity to freely generate a response to specific prompts as opposed to discriminating among several predefined responses (one of which is correct). In the latter case, i.e., multiple-choice questions, there is always the risk of learners picking the correct answer for the wrong reasons without the possibility of identifying such flawed knowledge. Open-ended responses eliminate such risks, reveal students' thought processes, document explanation of reasoning, and allow for the expression of creative and original correct responses, none of which is possible when using multiple-choice assessment instruments (Parmenter, 2009; Walstad & Becker, 1994). Therefore, freely generated student responses to a question, e.g., asked by an intelligent tutoring system (ITSs; Rus et al., 2013), or an essay prompt, e.g., as in the high-stake exam Scholastic Assessment Test (SAT), should be used in educational contexts because they provide a unique opportunity to assess students' knowledge and skills in an area.

The Need for and Challenges with Automated Assessment of Learner-Generated Natural Language Responses

The challenges (cost and effort) that arise from manually assessing open-ended student responses limit their use by educators. Automated methods to assess student-generated responses are fervently pursued to address the cost and effort challenges. It should be noted that automated methods have the additional advantage of systematically and consistently assessing student responses as compared to human raters. This chapter offers an overview of methods for automatically assessing students' freely generated answers.

The self-generation process, the key feature of open-ended assessment, offers unique opportunities and challenges for automated assessment. An effect of the self-generation aspect of open-ended responses, which is an advantage and a challenge at the same time, is their diversity along many quantitative and qualitative dimensions. For instance, free responses can vary in size from one word to a paragraph to a document. The challenging part is the fact that there needs to be a solution that can handle the entire variety of student responses, a tall order. Indeed, analyzing students' responses requires natural language techniques that can accommodate the endless variety of student input and assess such input accurately.

Another major challenge is that open-ended responses may be assessed in different ways that depend on the target domain and instructional goals. This makes it difficult to compare assessments. For example, in automated essay scoring the emphasis is more on how learners argue for their position with respect to an argument the essay prompt while in other tasks, such as conceptual physics problem solving, the emphasis is more on the content and accuracy of the solution articulated by the learner. This chapter provides an overview of the opportunities, challenges, and state-of-the-art solutions in the area of automated assessment of learner-generated natural language responses.

Automated Methods For Assessing Learner-Generated Natural Language Responses

We focus in this chapter on natural language open-student responses as opposed to responses that require, for instance, drawing a diagram. Assessing such natural language learner responses is a natural language understanding (NLU) task.

Computational approaches to NLU can be classified into three major categories: true-understanding, information extraction, and text-to-text similarity. In true understanding, the goal is to map language statements onto a deep semantic representation such as first-order logic (Moldovan & Rus, 2001; Rus 2002; Banarescu et al., 2013; Bender, Flickinger, Oepen, Packard & Copestake, 2015; Bos, 2015; Liang, Jordan & Klein, 2013). This approach relates language constructs to world and domain knowledge that is stored in a well-specified computational knowledge base (Lenat, 1995), and that ultimately enables inferencing. Inconsistency and contradictions can be automatically detected, revealing potential flaws in students' mental model of the target domain. Current state-of-the-art approaches that fall into this true-understanding category offer adequate solutions only in very limited contexts, i.e., toy-domains. These lack scalability and thus have limited use in real-world applications such as summarization or ITSs. Notable efforts in ITSs are by VanLehn and colleagues (Rosé, Gaydos, Hall, Roque & VanLehn, 2003; VanLehn, Jordan, Rosé, et al., 2002). Related efforts use less expressive logics like description logic (Zinn, Moore, Core, Varges, Porayska-Pomsta, 2003) in conjunction with the two other NLU approaches' categories discussed next.

Information extraction approaches use shallow processing to automatically detect in learners' free responses the presence of certain nuggets of information that represent key expected concepts or derived measures that could be used as predictors of student responses' correctness or overall quality. These approaches focus on text surface features or matching of exact words. They are most appropriate for item types such as fill-in-the-blank short answers where there is a limited range of expected correct responses.

Text-to-text (T2T) similarity approaches to textual semantic analysis avoid the hard task of true understanding by defining the meaning of a text based on its similarity to other texts, whose meaning is assumed to be known. Such methods are called benchmarking methods because they rely on a benchmark text, which is generated, checked, or annotated by experts, to identify the meaning of new, unseen texts. We focus in this chapter primarily on T2T approaches because they are the scalable and dominant practical approaches at this moment and probably for the foreseeable future. One may argue that some of the information extraction approaches fall under the category of T2T approaches because, for instance, identifying in student responses a number of expected concepts that are specified by an expert is equivalent with a T2T approach in which parts of student responses are compared to key expected concepts provided by experts. To make the distinction less ambiguous for such approaches, we call T2T approaches those that compare a full student response to a full benchmark or ideal response provided by an expert. If the expert only provides a set of key expected concepts, then such an approach falls under the information extraction category.

We also distinguish between two major categories of student responses: argumentative essays versus accurate-content responses, i.e., short essays. While these two types of essays are not mutually exclusive, nor are they inclusive of all essay types, these types of student responses are the result of two different major instructional goals. In argumentative essay scoring, there is an emphasis on *how* learners argue for their position with respect to the essay prompt, whereas in a task that scores conceptual physics problem solving the emphasis is on *what* is said, i.e., content/accuracy of the solution articulated by the learner. It should be noted that content is also considered in automated scoring of argumentative essays as, for instance, the essay must be "on topic", i.e., related to the essay prompt. Nevertheless, in this case factors such as vocabulary, grammaticality, and argumentation (the *how*) are emphasized. On the other hand, grammaticality and vocabulary size/richness are less important aspects to consider while assessing the correctness of physics problem solutions as compared to the conceptual correctness of the articulated solution. While grammaticality is not critical when assessing for correctness, it could be important because often content

knowledge cannot be fully explicated without correct expression. For instance, a response with a bad grammar will negatively impact automated syntactic parsing, which in turn will negatively impact the overall outcome of the automated assessment method. Bad grammar leads to bad parsing and so on, and it leads to error propagation throughout the automated assessment software pipeline. However, from our experience, we can testify that syntactic information does not add significant more value on top of other factors for predicting accuracy.

This chapter provides an overview of T2T opportunities, challenges, and state-of-the-art solutions. It is by no means a comprehensive review of previously published work in this area. Rather, it is a short summary of the area.

It is important to add one meta-comment regarding terminology before proceeding further with details of various T2T methods for automated assessment of learner-generated natural language responses. The authors of this contribution do not adhere to the so called “constructed student answers” label, which has been used by some researchers when referring to learner-generated natural language responses (Leacock & Chodorow, 2003; Magliano & Graesser, 2012). First, this label implies that students construct their answers from some predefined primitives/blocks, e.g., pieces of the response that might be available to them. This is not the case as students freely generate the answers. Second, the “constructed” terminology has its roots on the fact that experts construct a reference/model/ideal answer, which is then used to assess student responses. That is, the “constructed student answer” terminology links the nature of students’ answers to the evaluation method (Leacock & Chodorow, 2003; Magliano & Graesser, 2012), i.e., comparing the student answer to a constructed reference/model/ideal answer provided by experts. We believe that the way students generate the answer, e.g., freely composing the answer versus selecting it from a set of forced answer choices versus something else, should be the only aspect that should inform the choice of a label of what is being assessed. For instance, a freely composed student answer can be assessed by a true understanding method as opposed to a semantic similarity approach that relies on a reference answer or reference set of concepts.

Automated Essay Scoring

Automated essay scoring (AES) has become increasingly accepted with multiple systems available for implementing the scoring of writing for learning applications and ITSs (Shermis & Burstein, 2013). Studies of AES systems have shown that they can be as accurate as human scorers (e.g., Burstein, Chodorow & Leacock, 2004; Foltz, Laham & Landauer, 1999; Landauer, Laham & Foltz, 2001; Shermis & Hamner, 2012), can score on multiple traits of writing (Foltz et al., 2013), can provide accurate feedback on content (Beigman-Klebanov et al., 2014; Foltz, Gilliam & Kendall, 2000), and can score short responses (Higgins et al., 2014; Thurlow, et al., 2010). These systems are now also being used operationally in a number of high stakes assessments like General Educational Development (GED) and various state K–12 assessments (Williamson et al, 2010), placement tests like Pearson Test of English and ACCUPLACER (suite of tests that assess reading, writing, math, and computer skills), and for writing practice in systems like Criterion and WriteToLearn.

While there are differences among the various AES systems and the methods they employ, most share a common approach. This approach can be simply described as 1) represent a student’s writing quantitatively as a set of features and 2) determine how to weigh and combine the features to best characterize the quality of the student writing. The features are quantifiable natural language processing features that measure aspects such as the student’s expression and organization of words and sentences, the student’s knowledge of the content of the domain, the quality of the student’s reasoning, and the student’s skills in language use, grammar, and the mechanics of writing. In creating these features, it is critical that the computational measures extract aspects of student performance that are relevant to the constructs for the competencies of

interest (e.g., Hearst, 2000; Williamson, Xi & Breyer, 2012). For example, and explicated in greater detail in the following, features that measure the semantic content of student writing are used as measures of the quality of a student's domain knowledge. A measure of the type and quality of words used by a student provides a valid measure of lexical sophistication. However, measures that simply count the number of words in an essay, although it may be highly correlated with human scores for essays, do not provide a valid measure of writing sophistication.

To score a student-generated essay, multiple language features are typically measured and combined to provide a score or feedback. Combining and weighing the derived features is then performed by learning to associate the features with examples of student performance. These examples of performance can include samples of student responses that have been pre-scored by human raters or examples of ideal responses created or selected by experts. Machine learning (e.g., regression, Bayesian, classification) techniques are then used to build a model that weighs the features in relationship to the examples of performance while maximizing predictive performance and generalizability. The details of these approaches are beyond the scope of this chapter, but are covered in detail in Shermis and Burstein, (2013). However, we do describe important considerations in the features and methods used for assessing student-generated essays, most particularly with respect to content accuracy.

The recent Automated Student Assessment Prize (ASAP), funded by the Hewlett Foundation, revealed that current AES technologies rival human grader's performance (Shermis & Hamner, 2012; Shermis, 2014), although there are debatable aspects of the used methodology (Perelman, 2013). The ASAP exercise, which drew participation from seven commercial organizations and one academic lab, has shown that the dominant approach is a form of semantic similarity in which new, to-be-scored essays are compared to human-graded essays whose quality is already known.

Accurate-Content Essay Scoring

When assessing learners' responses for content accuracy or correctness, the most widely used approach is semantic similarity. As already mentioned, such an approach involves experts providing one or more ideal responses to specific prompts or sets of student responses that are pre-scored by experts. New student responses to the same prompts are subsequently compared to these ideal or benchmark responses. The main advantage of the semantic similarity approach is the circumvention of the need to acquire and automatically use world and domain knowledge explicitly, which are required for a true understanding of learners' natural language responses. It is assumed and hoped that the ideal response provided by experts contains enough cues that allows a simple or augmented semantic similarity process to make a good enough judgment on the correctness of students' responses.

While a semantic similarity approach still requires experts to generate ideal/benchmark responses, or score previously collected student responses, this manual step leads, overall, to a more scalable and cost-effective approach than the true understanding approaches. Consequently, the semantic similarity approach currently dominates.

Before presenting more details about such semantic similarity approaches to assessing the accuracy/correctness of learner responses, we present key issues with automatically assessing students' answers using a semantic similarity approach. Given the focus of this volume, we focus on student answers that need to be assessed in the context of intelligent interactive systems, namely, ITSs. In these systems, learners are prompted to provide answers and these answers must be assessed in real time to provide appropriate feedback and trigger appropriate instructional strategies that maximize learning.

Key Challenges

There are many challenges when it comes to automatically assessment of students' open-ended responses. We start by enumerating some of these challenges.

Spelling and Grammatical Errors. Students' responses often contain spelling and grammatical errors. Depending on various learners' verbal and writing abilities, they will generate responses that vary in compositional quality. The good news is that reasonably good spelling and grammar correction software tools are available, at least for English. Furthermore, since the focus on content-accuracy assessment is often on content, issues such as spelling and grammatical errors are not a major conceptual hurdle to overcome. That being said, there are pragmatic implications due to the current state of natural language technologies, e.g., modest syntactic parsing accuracy, as mentioned earlier.

Size Variability. There is great variability in student responses when it comes to size. For instance, student responses can be a single word, a phrase or chunk, a segment of a sentence such as a clause or several clauses, a complete sentence, a paragraph/short-essay or even a full essay. One needs to develop an approach that can handle student responses of various granularity or, alternatively, develop a separate approach for each of the mentioned granularity levels. Some approaches can be easily extended from a word to a sentence level but not to multiple-sentences/paragraph level. Other approaches scale well from words to sentences to paragraphs at the expense of disregarding information that is important for precise assessments (Rus, Banjade & Lintean, 2014).

Dialogue Utterances. Specific challenges arise when assessing learners' language in a multi-turn dialogue context such as in dialogue-based ITSs. For example, elliptical responses similar to response #5 in Table 1 are quite frequent. In spoken dialogue, the challenges are even greater due to peculiarities of spoken dialogue, which we address later in the spoken versus written input section.

Heavily Contextualized. In many situations students' responses are highly contextualized. For example, all the student answers shown in Table 1 are in the context of one particular physics problem that was used in an experiment with high school students who learned physics from a state-of-the-art ITS called DeepTutor (Rus, D'Mello, Hu. & Graesser, 2013). All these student responses must therefore be assessed while accounting for the physics problem, which is the focus of the dialogue. Furthermore, the responses needed to be interpreted in the context of the full dialogue history up to the point where the response was generated. In general, the responses in Table 1 are targeted responses to a previous tutor question, which is yet another contextual dimension.

Table 1. Examples of actual high school student answers showing the diversity of student responses to various prompts from the state-of-the-art intelligent tutoring system DeepTutor.

ID	Student Responses
1	The force exerted by gravity and tension of the rope are equal.
2	These forces balance each other.
3	The tension is equal to the force of gravity.
4	They are equal.
5	Equal.
7	The tension in the rope is greater than the downward force of gravity.
8	The tension in the rope is greater than gravity in order to raise the child upward.
9	They are equal and opposite in direction.
10	The tension in the rope is equal to the mass of boy times gravity. Newton's second law states the force is equal to mass times acceleration. In this case, the tension is the force. Gravity is the acceleration.

Textual responses vs. multi-modal. This chapter focuses on textual student responses but multimodal responses that contain text, diagrams, and other non-textual products, e.g., a diagram, are often generated and need to be assessed. There are specific challenges in these cases such as aligning the textual and non-textual elements to generate a more complete model of the student response. Even when student responses are pure textual, there may be reference to non-textual elements provided in context, e.g., a physics problem usually has a picture attached to it describing a visual rendering of the problem scenario. Students may refer to the provided picture instead of simply focusing on the concepts mentioned in the textual description of the problem. For instance, students may simply say “*The truck is pushing the car to the right*” when no explicit spatial relationship was mentioned in the problem description or the previous dialogue with the ITS; such spatial information, although irrelevant to answering the main question of the problem, is conveyed through the accompanying image depicting a truck pushing a car.

Core Linguistics Issues. Additional key linguistic issues often need to be addressed for a more comprehensive solution even when a semantic similarity approach is being used. These key linguistics tasks include coreference and anaphora resolution as students often refer to entities mentioned earlier in the dialogue and problem description using, for instance, pronouns (Niraula, Rus, Stefanescu, 2013; Niraula, Rus, Banjade, Stefanescu, Baggett, and Morgan, 2014), negation (Banjade, Niraula, and Rus, 2016), and synonymy (different words with same meaning) and polysemy (many meanings of the same word). Some of the semantic similarity approaches we summarize do include simple and sophisticated solutions for word sense disambiguation to address, for instance, polysemy. Semantic methods such as Latent Semantic Analysis (LSA; Landauer, Foltz & Laham, 1998) handle synonymy indirectly by mapping words into a semantic space where synonym words will be close to each other.

Written vs. Spoken Input. Students’ responses can be typed or spoken. Spoken language has two main challenges: register versus technical. There are specific phenomena that are more prevalent in spoken language than written language. For example, there are disfluencies such as stop-restart segments where a speaker starts an utterance and suddenly stops before finishing it to restart and utter a replacement. Concrete examples of such disfluencies are fillers (“uh”, “um”), which are dialogue specific particles with no particular content value, false starts/re-starts (“I really wanted ...”), repetitions (“I would like a coffee a hot coffee”), and corrections (“I would like a an apple”).

Compositional Approaches

We focus in this section on compositional approaches to addressing the task of automated assessment of student open responses. The *principle of compositionality* states that the meaning of a text can be determined by the meaning of its constituents and the rules used to combine them (Lintean & Rus, 2012). The major steps in compositional approaches are outlined as follows:

- *Step 1. Derive a word-level meaning representation.* This might include some necessary preprocessing of the input text, which could include the detection of other constructs such as a noun phrase or a verb phrase.
- *Step 2. Discourse level processing.* This includes resolution of content references, such as pronouns to their referents, to maximize the outcome of the next step, i.e., the alignment step.
- *Step 3. Alignment.* Align words (or other constituents) across the paired student vs. ideal responses and combine the word-level meaning representations. The alignment could be based on lexical information, i.e., words and their meanings, and also take into account relations among words such as syntactic relations. Other contextual elements such as the nearby words (within a window of, say, three words before or three words after) or directly related words via a syntactic dependency could be considered. The alignment could also include explicit negation particles.

- *Step 4. Report an overall, usually normalized, similarity score.* The resulting score could be sanctioned by extra semantic elements such as negation focus and scope. For instance, Rus and Graesser (2006) altered the alignment score using a negation term that accounted for single or double negation.
- *Step 5. Map the similarity score into qualitative decisions.* After obtaining the similarity score a qualitative decision is being derived such as the student response is correct or the student response is incorrect.

Word-level and Text-level Meaning Representations

We focus in this section on the first step of the semantic similarity procedure outlined previously and which consists of deriving meaning representations of texts, a key issue in NLU. As previously stated, T2T approaches define the meaning of a text based on its similarity to other texts.

Perhaps the simplest approach is to represent texts as sets of words and as a similarity metric the set intersection of the words of the two texts under consideration, i.e., $U \cap V$ for texts U and V . This intersection can be normalized by the cardinality or union of U and V to yield a variety of set-based metrics initially proposed by Paul Jaccard and known as the Jaccard similarity coefficient. Set-based metrics have the advantage that the calculation of similarity between texts is fairly direct (Manning & Schütze, 1999). In this family of “set-of-words” approaches, words are not given any particular kind of representation but texts are treated as sets of words. As we explain later, even for such methods of set-based textual semantic representations and similarity approaches, there is an underlying 1-of- N vector representation with binary weights (as opposed to representations based on distributional properties of words derived from large collections of natural language texts, which are explained next).

A more sophisticated family of approaches is based on the distributional vector space representation in which the meaning of a word is represented as a vector in a multi-dimensional space. The dimensionality of the space varies from one vector space representation to another. Furthermore, these distributional vector space approaches rely on the distributional hypothesis according to which the meaning of a word is defined by the company it keeps. That is, they derive the vector representation of words starting with a word-to-word co-occurrence analysis over a large collection of texts, as detailed later. Originally conceived for information retrieval, where the purpose is to match a textual query with the most relevant or similar document in a large collection of documents (Salton, Wong & Yang, 1975), distributional vector space approaches are perhaps the most widely used for T2T similarity.

As already mentioned, distributional vector representations rely on a statistical analysis of word co-occurrences in large collections of natural language texts, i.e., a corpus such as Wikipedia articles (Ștefănescu, Banjade & Rus, 2014). A typical and simple co-occurrence analysis consists of generating a term-by-document matrix where each row vector represents a particular word or term and the columns represent the documents. Thus, each row captures statistical distributional information of a word across documents and each column represents distributional information of the words in a document, i.e., co-occurrence information among words within a single document. Furthermore, this statistical distributional information of a word/term across documents and of word/term co-occurrence within a document is captured in an elegant algebraic format of a term-by-document matrix. This representation then enables the use of algebraic operations, an important attraction of this representation, serving different purposes. For instance, multiplying the term-by-document matrix by its transpose results in a term-by-term co-occurrence matrix where each cell provides a quantitative summary of how two words/terms co-occur with each other across all documents in the collection.

To construct the term-by-document matrix, many different approaches may be used. The original information retrieval approach is a frequency approach where words are counted over documents. Thus, a given cell_{ij} of the matrix represents the number of times that word_i appears in document_j . Because not all words occur in all documents, the row vectors are necessarily sparse. The similarity between words can be computed based on the similarity of their vectors, using metrics like dot product or cosine, i.e., normalized dot-product, to eliminate bias toward longer documents. Furthermore these word vectors have the property that they may be added together, allowing one to obtain a vector of a chunk of text, e.g., a phrase or sentence or paragraph, by adding the vectors for words in the text. This additive process to obtain the meaning of a larger text from its individual words is a simple instantiation of the meaning compositionality principle (the simple co-occurrence in the same text is, in this case, the rule-to-combine or “relationship” specified in the definition of the principle of compositionality mentioned earlier). The resulting vectors can be compared with any other vector using the same vector similarity metrics.

Although this basic term-by-document structure is shared by all members of the vector space family, variations are wide ranging. For example, words can be preprocessed before computing their frequencies using techniques like stemming (Rus, Banjade & Lintean, 2014). Rus, Banjade, and Lintean (2014) discuss in detail the implication of preprocessing steps on the performance of semantic processing approaches and show that such preprocessing steps have a much more significant impact than recognized by the research community (it is important to note that they identified and considered 1,152 combinations of preprocessing steps in their study). Raw counts may also be transformed in a variety of ways using information about the cell, its row, or its column. For example, the presence of a word in a document is typically more important than the number of times the word appeared, leading to word frequency normalization like log scaling of cell counts, and words that appear in many documents are less diagnostic of meaning and so can be down-weighted using row weighting schemes like inverse document frequency; similarly approaches based on information theory can be used to weight cell counts based on word (row) or column (document) distributions (Church & Hanks, 1989; Dumais, 1991; Manning & Schütze, 1999).

The set-based similarity approaches that we mentioned earlier can be regarded as using an underlying vector representation with binary weights: the word is present, corresponding to a value of 1, or not, corresponding to a value of 0. The dot product between such binary vectors corresponding to two texts results in the cardinality/size of the intersection of the underlying sets of words, i.e., the number of common words in the two texts. When using raw frequency as the weights, the result is a similar vector representation, which regards the texts as bag of words (multiple occurrences of the same word are accounted for) as opposed to sets of words (multiple occurrences of the same word only count once). In fact, we can think of a vector representation for individual words even if an explicit one is not derived from a corpus. Each word can be thought of as having a 1-of-N representation, where N is the size of the vector, i.e., the number of entries or dimensionality (and equal to the vocabulary size of the two texts), and only one cell or entry that corresponds to dimension of the target word has a weight of 1 (all other entries being zero).

Finally, some approaches transform the matrix into another matrix using matrix factorization techniques. In particular, the use of singular value decomposition to factorize the matrix is strongly associated with the vector space approach known as latent semantic analysis (LSA; Landuaer, Foltz & Laham, 1998; Landauer, McNamara, Dennis & Kintsch, 2007). The computational advantage of LSA is that it represents that meaning of words using a reduced dimensionality space (300–500 dimensions) leading to fast computations of similarity scores, e.g., based on cosine similarity. A study comparing the effect of various local and global weighting schemes in conjunction with LSA was described by Lintean, Moldovan, Rus, and McNamara (2010).

Although the basic approach outlined previously seems very sensible from an information retrieval perspective focused on documents, from a generic text similarity perspective, aspects of this approach are

somewhat arbitrary. For example, documents are not defined to have any particular length, leading to questions about whether there is an optimum length for documents or whether it matters if document lengths across a matrix are highly variable. Likewise, it is useful to have words as the unit of co-occurrence analysis and representation but in many applications of natural language processing sequences of words called *n*-grams have been used with great success. Considerations such as these have led to proposals for abstracting from the word by document matrix to a feature by context matrix (Olney, 2009). In these matrices, the concern is counting a particular feature within a given context, where both context and feature can be arbitrarily defined. For example, a square matrix could be defined with rows and columns equal to the number of words in the collection, so a $cell_{ij}$ might represent the number of times word_i occurs after word_j. Likewise a row could be a word and $cell_{ij}$ might represent the number of times word_i is the syntactic head of word_j.

Taking a feature by context perspective is useful when considering approaches where there is no document or the document is part of a larger interconnected structure. For example, the Correlated Occurrence Analogue to Lexical Semantics model (COALS; Rohde, Gonnerman & Plaut, 2005; Olney, Dale & D’Mello, 2012) implements a sliding window strategy. Similar to the previous example, the resulting matrix is square, but the cell counts are based on a symmetric window around the target word. The word at the center of that window corresponds to the row of the matrix, and the incremented columns are the other words within the window; however, the increment is inversely proportional to the distance between them and the target word. The matrix is normalized using phi correlation and then optionally transformed using singular value decomposition.

Structured texts, like Wikipedia, have inspired structure-specific approaches defining features and contexts. Explicit Semantic Analysis (ESA) uses the article structure of Wikipedia to determine contexts, and thus a $cell_{ij}$ represents the number of times word_i occurs in article_j. ESA further weights the cells using log weighting of word counts and inverse document frequency. Another approach that uses the structure of Wikipedia to define both features and contexts is Wikipedia Link Measure (WLM; Milne & Witten, 2008). WLM uses articles as features and links to them from other Wikipedia pages as contexts. To compare the similarity of two words, WLM associates them with articles and then defines a metric over the shared links to those pages (in-links) and shared links from those pages (out-links) to determine similarity. Although the original definition of WLM is more set theoretic based on the graph structure of links, it trivially maps to a vector space approach where links are columns in a matrix (Olney et al., 2012).

One other word meaning representation worth noting is Latent Dirichlet Allocation (LDA; Blei, Ng, and Jordan, 2003). LDA is a probabilistic, unsupervised method that models documents as distributions over topics and topics as distributions over words in the vocabulary. Each word has a certain contribution to a topic. That is, each word can be represented as a vector whose dimensionality equals the number of latent topics and the weights along each dimension correspond to the word contributions to each topic as derived by the LDA method. Based on these distributions and contributions word-to-word semantic similarity and text-to-text semantic similarity measures can be defined (Rus, Niraula & Banjade, 2013). It should be noted that LDA has a conceptual advantage over LSA because LDA explicitly represents different meanings of words, i.e., it captures polysemy. In LDA, each topic is a set of words that together define a meaning or concept, which corresponds to a specific sense of the various words sharing this meaning. Thus in LDA, each topic a word belongs to can be regarded as one of its senses. On the other hand, LSA has a unique representation for a word. That is, all meanings of a word are represented by the same LSA vector making it impossible to distinguish among the various senses of the word directly from the representation. Some argue that the LSA vector represents that dominant meaning of a word while others believe the LSA vector represents an average of all the meanings of the word. The bottom line is that there is a lack of explicit account for the various senses of a word in LSA.

Yet another major trend in the area of distributional vector-based representations, is the new category of representations derived using deep neural networks. A typical example in this category is the word2vec

representation developed by Mikolov and colleagues (2013). They trained a recursive neural network with local context (continuous n-gram or skip-gram) using a process in which the input word vectors are recursively adjusted until the target context words are accurately predicted given the input word. These representations were shown to capture syntactic and lexical semantic regularities, have superior compositionality properties, and enable precise analogical reasoning using simple vector algebra (Mikolov, Chen, Corrado & Dean, 2013; Pennington, Socher & Manning, 2014).

A Categorization of Compositional based Semantic Similarity Approaches

Once a word meaning representation is being settled upon, the meaning of larger texts can be derived compositionally in many different ways. We distinguish among the following major categories of compositional approaches.

Simple Additive Approaches

Simple additive approaches generate an overall semantic similarity score by applying simple operations, such as addition and average matching, to word level representations. Two examples of such simple additive approaches are the vector addition and word-average matching addition (Fernando & Stevenson, 2008). In the vector addition approach, an overall vector for each text is computed by simply adding up the vector representations of individual words. Then a vector similarity metric is applied on the resulting vectors of the two target texts. In the word-average matching addition approach, for each word in one text an average semantic similarity score is computed with all the words in the other text. To compute this average similarity score for a single word, the word is matched or paired with every single word in the other text and a similarity metric is computed between the corresponding vector representations of the paired words. The average of all those word-to-word similarity scores is then taken.

Alignment-Based Approaches

In this category, the approaches have a distinguished feature of aligning first the words in one text to at most one word in the other text. From a student answer assessment point of view, performing the alignment first has the great advantage of detecting which main concepts the student articulated and which ones are missing from the response. Detecting the articulated and missing concepts, in turn, enables automated generation of feedback and informs the dialogue/interaction planning component of the underlying ITS.

The alignment can be based on simple lexical matching (using the words themselves) in which case we end up with a simple lexical overlap score, i.e., the number of common words between the two texts. This simple lexical overlap score is computed after applying the various preprocessing steps chosen by the designer such as lemmatization and can be normalized by the average length or the maximum length or some other normalizing factor (see Lintean, Moldovan, Rus & McNamara, 2010; Rus, Banjade & Lintean, 2014). The two texts can be treated as sets or as bags. In the latter case, words that match can be weighted by their frequencies in the two texts to account for multiple occurrences.

A more advanced approach, called lexical similarity alignment, relies on word-to-word (w2w), or lexical, semantic similarity measures to find the best alignment between a word in one text and at most one word in the other text. We distinguish between two types of lexical similarity alignment approaches: *greedy lexical similarity alignment* and *optimal lexical similarity alignment* that only relies on lexical similarities between words but not on their relationship (e.g., syntactic and/or deeper semantic relations). The greedy approach simply aligns a word in one text with the word in the other text that leads to the highest w2w similarity score according to some w2w similarity measures.

The optimum assignment between words in one text, T1, and words in another text, T2, can be posed as a combinatorial optimization problem. That is, the goal is to find a permutation π for which $\sum_{k=1}^n \Theta(v_i, w_{\pi(i)})$ is maximum, where Θ denotes any word-to-word similarity measure, and v and w are words from the texts T1 and T2, respectively. This formulation of the T2T similarity problem is in fact the famous job assignment problem for which an algorithm, the Kuhn-Munkres or Hungarian method (Kuhn, 1955), has been proposed and which can find a solution in polynomial time.

The assignment problem only focuses on optimally matching words in one sentence S to words in the other sentence T based only on how the words in S match the words in T. As briefly mentioned, it does not account for interdependencies among words in S or among words in T. A solution that simultaneously accounts for such inter-dependencies, thus capturing the context of each word in their corresponding sentences, has been proposed by Lintean and Rus (2015).

The *optimal lexico-relational alignment* method aims at finding an optimal global assignment of words in one sentence (e.g., a student response) to words in the other sentence (e.g., the expert answer) based on their w2w similarity, while simultaneously maximizing the match between the syntactic dependencies. Accounting for the syntactic dependencies among words is the primary advantage of the quadratic assignment problem (QAP; Koopmans-Beckmann, 1957) formulation versus the job-assignment formulation of the student response assessment task (Rus & Lintean, 2012).

The formulation of the QAP problem for textual semantic similarity proposed by Lintean and Rus (2015) is to maximize the objective function QAP (see below), where matrix F and D describe dependencies between words in one sentence and the other, respectively, while B captures the w2w similarity between words across the two texts. Also, they weighted each term resulting in the following formulation:

$$\max QAP(F, D, B) = \alpha \sum_{i=1}^n \sum_{j=1}^n f_{i,j} d_{\pi(i)\pi(j)} + (1 - \alpha) \sum_{i=1}^n b_{i,\pi(i)}$$

The $f_{i,j}$ term quantifies the (syntactic or semantic or of other nature) relation between words i and j in text A which are mapped to words $\pi(i)$ and $\pi(j)$ in text B, respectively. The distance $d_{\pi(i)\pi(j)}$ quantifies the semantic relation between words $\pi(i)$ and $\pi(j)$. For words i and j that have a direct syntactic dependency relation, i.e., an explicit syntactic relation among two words such as subject or direct object, the “flow” $f_{i,j}$ is set to 1 (0 – if not direct relation). Similarly, the distance $d_{\pi(i)\pi(j)}$ between words $\pi(i)$ and $\pi(j)$ is set to 1 in case there is a direct dependency relation among them and 0 otherwise.

A brute force solution to the QAP problem, which would generate all possible mappings from words in a sentence to words in the other sentence, is not feasible because the solution space is too large. For example, when considering all possible pairings of words between sentence A, of size n , and sentence B of size m , where $n < m$, and there are no limitations on the type of pairings that can be made, there are $m!/(m-n)!$ possible solutions. For sentences of average size $n = m = 20$ words, there are $2.4 * 10^{18}$ possible pairings. An branch-and-bound algorithm was proposed by Lintean and Rus (2015) to efficiently explore the solution space in search for the optimal solution.

Interpretable alignment based approaches perform first an alignment between words in one text versus words in the other texts while at the same time identifying semantic labels for aligned words (Banjade, Maharjan, Gautam, and Rus, 2016). The advantage of adding semantic relationships between the aligned words is that an explanation for the alignment can be provided based on these w2w semantic relationships. The set of semantic relationships used were proposed as part of the interpretable Semantic Textual Similarity (iSTS) task (Agirre et al., 2016), organized by SemEval – the leading semantic evaluation forum, and includes: EQUI (semantically equivalent), OPPO (opposite in meaning), SPE (one chunk is more specific

than other), SIMI (similar meanings, but not EQUI, OPPO, SPE), REL (related meanings, but not SIMI, EQUI, OPPO, SPE), and NOALI (has no corresponding chunk in the other sentence). It should be noted that we presented alignment based methods focusing on words. Other units of analysis can be used such as chunks (Stefanescu, Banjade & Rus, 2014a).

Resources

Aside from a quantitative and qualitative outcome for T2T tasks, there recently has been a push to offer an explanation of the final T2T outcome. To this end, several resources and efforts have been reported as explained below. Additionally, we list some other relevant resources that have been developed and released publicly in order to foster research in this area.

The SEMILAR Corpus. Rus and colleagues (2012) developed the SEMILAR corpus, which is the richest in terms of annotations as besides holistic judgments of paraphrase they provide several word level similarity and alignment judgments. The corpus includes a total of 12,560 expert-annotated relations for a greedy word-matching procedure and 15,692 relations for an optimal alignment procedure.

The Student Response Analysis (SRA) Corpus. The SRA corpus (Dzikovska et al., 2013) consists of student answer-expert answer pairs collected from two ITSs. Both student answers and expert answers were related to specific tutorial questions from different science domains. There are 56 questions and 3,000 student answers from the so-called BEETLE corpus as well as 197 assessment questions and 10,000 answers from the SciBank corpus. These pairs were annotated using a combination of heuristics and manual annotation. They used a 5-way annotation as opposed to the typical 2-way annotation.

The User Language Paraphrase Corpus (ULPC; McCarthy and McNamara 2008). ULPC contains pairs of target sentence/student response texts. The student responses were collected from experiments with the ITS iSTART. Students were shown individual sentences collected from biology textbooks and asked to paraphrase them. These pairs have been evaluated by expert human raters along 10 dimensions of paraphrase characteristics. The “paraphrase quality bin” dimension measures the paraphrase quality between the target-sentence and the student response on a binary scale. From a total of 1,998 pairs, 1,436 (71%) were classified by experts as being paraphrases. A quarter of the corpus is set aside as test data. The average words per sentence is 15.

DeepTutor Anaphora Resolution Annotated (DARE): The DARE corpus (Niraula, Rus, Banjade, Stefanescu, Baggett, and Morgan, 2014) is an annotated data set focusing on pronoun resolution in tutorial dialogue. Although data sets for general purpose anaphora resolution exist, they are not suitable for dialogue-based ITSs, which is the reason the DARE corpus was created. The DARE corpus consists of 1,000 annotated pronoun instances collected from conversations between high school students and the ITS DeepTutor. The data set is publicly available.

DeepTutor Tutorial Dialogue Negation Annotated (DT-Neg; Banjade and Rus, 2016) Corpus. Negation is found more frequently in dialogue than typical written texts, e.g., literary texts. Furthermore, the scope and focus of negation depends on context in dialogues more so than in other forms of texts. Existing negation data sets have focused on non-dialogue texts such as literary texts where the scope and focus of negation is normally present within the same sentence where the negation cue is located and therefore are not the most appropriate to inform the development of negation handling algorithms for dialogue-based systems. The DT-Neg corpus contains texts extracted from tutorial dialogues where students interacted with the ITS to solve conceptual physics problems. The DT-Neg corpus contains 1,088 annotated negations in student responses with scope and focus marked based on the context of the dialogue.

DeepTutor Student Answer Grading Context-aware Annotated (DT-Grade; Banjade, Maharjan, Niraula, Gautam, Samei & Rus, 2016) *Corpus*. The DT-Grade corpus consists of short constructed answers extracted from tutorial dialogues between students and the DeepTutor ITS and annotated for their correctness in the given context and whether the contextual information was useful. The data set contains 900 answers of which about 25% required contextual information to properly interpret.

Semantic Similarity Toolkit (SEMILAR; Rus, Lintean, Banjade, Niraula & Stefanescu, 2013). The SEMILAR software package offers users (researchers, practitioners, and developers) easy access to fully-implemented semantic similarity methods in one place through both a graphical user interface (GUI)-based interface and a software library. Besides productivity advantages, SEMILAR provides a framework for the systematic comparison of various semantic similarity methods. The automated methods offered by SEMILAR range from simple lexical overlap methods to methods that rely on w2w similarity metrics to more sophisticated methods that rely on fully unsupervised methods to derive the meaning of words and sentences such as LSA and LDA to kernel-based methods for assessing similarity.

Conclusions and Recommendations for Future Research

Given the importance of student-generated open responses in educational context, which, we argue, are the only assessment modality that leads to true assessment because are the only assessment modality that reveals students' true mental model, future educational technologies including the Generalized Intelligent Framework for Tutoring (GIFT; Sottolare, Brawner, Goldberg & Holden, 2012; Sottolare, Brawner, Sinatra & Johnston) should include open-ended assessment items and corresponding facilities that enable the automated assessment of such open-ended student responses.

True assessment is necessary to infer an accurate picture of students' mastery level, which in turn is paramount for triggering appropriate feedback and instructional strategies and ultimately the effective and efficiency of the underlying educational technology. Considering the early stage nature of the assessment module in the educational processing pipeline and therefore the positive or negative cascading effect it may have on the downstream modules (learner model, feedback, strategies, and outcome, e.g., learning) the importance of automated assessment of open-ended learner responses cannot be overstated.

Assessing students' open-ended responses is complex and requires a multitude of factors to be considered, as illustrated in this contribution. However, this complexity is surmountable and there has been tremendous progress in terms of advanced methods and resources that have been developed and publicly released.

Acknowledgments

The research was supported in part by the Institute for Education Sciences under award R305A100875, under Cooperative Agreement W911NF-12-2-003 between U.S. Army Research Laboratory (ARL) and the University of Memphis, Office of Naval Research (N00014-00-1-0600, N00014-15-P-1184; N00014-12-C-0643; N00014-16-C-3027) and the National Science Foundation Data Infrastructure Building Blocks program (ACI-1443068). Any opinions, findings, and conclusions expressed are solely the authors'.

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